

Algorithms: Endsem Exam

28 Apr 2026

Total Marks: 40

Time: 2 hrs 30 min

1. Suppose that you have k sorted arrays, each with n elements, and you want to combine them into a single sorted array of kn elements.

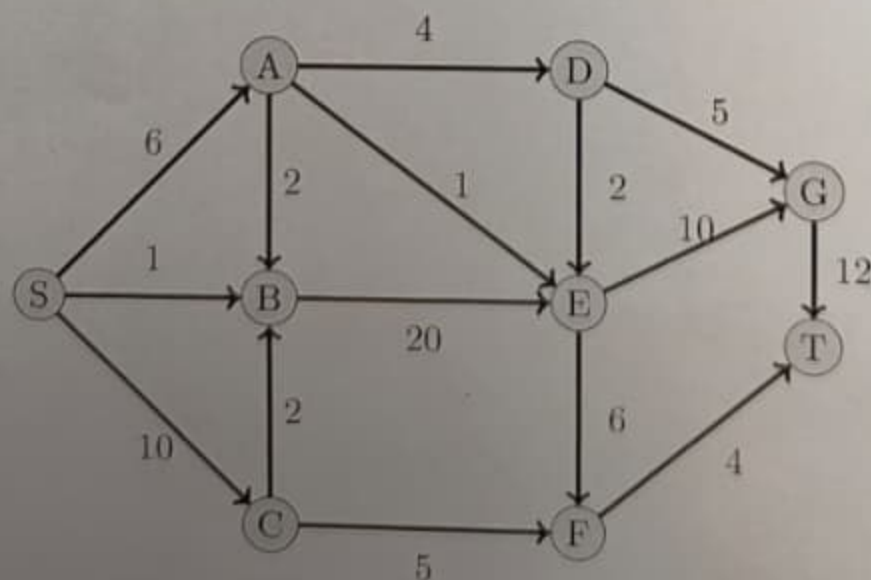
(a) Consider a sequential merging strategy. Merge the first two arrays into a new array, then the third array with this new array, then the fourth array and so on. What is the time complexity of this algorithm, in terms of k and n ? [2 marks]

(b) Give a more efficient solution to this problem, than the one in (a). [4 marks]

2. Imagine that you have H hours of leisure to spend in your vacation. There are several activities which you have planned and which increase your (short-term) happiness up to an extent. We'll assume that this is measurable in terms of happiness points. For example, there may be an activity which gives you 10 points after 1 hour, 20 points with 2 hours, 10 points with 3 hours and zero points for 4 or more hours. Your goal is to schedule your activities to maximize your happiness score.

The input is a positive integer H and a matrix $A_{(i,j)}$ where $1 \leq i \leq n$ and $0 \leq j \leq H$. Here, $A_{i,j}$ denotes the points you gain by spending j hours on the i th activity. Your goal is to divide your budget of H hours across activities to maximize the total points that you get. That is, find $h_1, h_2, \dots, h_n$ such that $h_1 + \dots + h_n = H$ with the maximum value of $\sum_{i=1}^n A(i, h_i)$. Describe and analyze an efficient algorithm for this problem. [6 marks]

3. For the following graph, with edge capacities as shown, find the maximum flow from S to T using the Edmonds-Karp algorithm. [6 marks]



4. You are given a directed graph $G = (V, E)$ with weights on both vertices and edges. The weight of a path is defined to be the sum of the weights of the edges on the path and the weights of the vertices on the path (including the endpoints). Assuming that all weights are positive, describe an efficient algorithm to find the minimum weight path from a given source vertex s to all other vertices in the graph. [6 marks]

5. Donald suggests the following algorithm for finding the shortest path from node s to node t in a directed graph with some negative edges: add a large constant to each edge weight so that all the weights become positive, then run Dijkstra's algorithm starting at node s , and return the shortest path found to node t .

Is this a valid method? Prove its correctness or disprove it by giving a counterexample.

[5 marks]

6. Given a directed graph $G = (V, E)$, explain how to create another graph $G' = (V, E')$ such that (i) G' has the same strongly connected components as G , (ii) G' has the same component graph as G , and (iii) E' is as small as possible. For example if G consists of ~~the~~^{six} vertices a, b, c, d, e, f and edges $(a, b), (b, c), (c, d), (d, a), (a, c), (c, e), (d, e), (e, f), (f, e)$, then G' can consist of the edge set $E' = (a, b), (b, c), (c, d), (d, a), (a, c), (c, e), (e, f), (f, e)$. [5 marks]

7. Given a graph $G = (V, E)$, we say that a subset $S \subseteq V$ of vertices is a **covering set** if for every $v \in V$, at least one of the following holds:

(i) $v \in S$;

(ii) There exists $w \in S$ such that $\{v, w\} \in E$.

In other words, every vertex must belong to S or have a neighbor in S (or both).

For example, if G has four vertices all of which are pairwise adjacent, then any of the vertices by itself is a covering set, thus such a graph has a covering set of size one.

Given a tree as input, describe an efficient algorithm to compute a minimum size covering set for the tree. [6 marks]